

Math 300 Quiz 6 Spring 2024

1. (8 pts) Use Cauchy's definition to show $\lim_{x \rightarrow 1} (3x^2 - 7) = -4$

2. (4 pts) Use the definition for the derivative of $f(x)$ to show that $D_x[x^2 + 3] = 2x$.

$$f'(x) =$$

Over $\rightarrow$

3. (4 pts) Given $\lim_{x \rightarrow 3} f(x) = 5$ and $\lim_{x \rightarrow 3} g(x) = 2$.

$$\lim_{x \rightarrow 3} [2f(x) + \sqrt{g(x)} + 10] =$$

4. (4 pts) Given $f(2) = 7$, and $f'(2) = -5$ what is $h'(2)$ if $h(x) = \frac{f(x)}{x}$?